

Monitoring the Power Consumption of Home Appliances using an IoT-based SCADA System

Sugeng Priyanto

Department of Electrical Engineering
National Institute of Technology (ITN)
Malang
Malang, Indonesia
sugengp.priyanto@gmail.com

Aryuanto Soetedjo

Department of Electrical Engineering
National Institute of Technology (ITN)
Malang
Malang, Indonesia
aryuanto@lecturer.itn.ac.id

Irrine Budi Sulistiawati

Department of Electrical Engineering
National Institute of Technology (ITN)
Malang
Malang, Indonesia
irrine@lecturer.itn.ac.id

Abstract— A home energy management system (HEMS) is part of a Smart Grid, a modern technology in the electricity system. The HEMS provides efficient energy consumption at home. This paper presents a development of the IoT-based Supervisory Control and Data Acquisition (SCADA) system for monitoring the power consumption of home appliances. The SCADA application is implemented on small-size Human Machine Interface (HMI) devices and communicates with the power measurement devices wirelessly using the Modbus TCP protocol. The developed system is tested at home to monitor the power consumption of the refrigerator, washing machine, rice cooker, and fan in real time. The experiment results show that the SCADA monitoring system runs continuously during the week. Further, the recorded home appliances' power consumption profiles give enough information about the appliances' characteristics.

Keywords—home appliances, monitoring system, power consumption, IoT-based SCADA, HMI, Modbus TCP

I. INTRODUCTION

Managing the power consumption of home appliances plays an essential role in the climate change issue. A system to monitor load power consumption helps users know how to save energy [1]. There are two methods for monitoring power consumption [1]. The first is to monitor individual loads using a power meter called intrusive appliance load monitoring (IALM). The second is a nonintrusive appliance load monitoring (NIALM) system or energy disaggregation, in which the power meter is only one and is located near the energy meter.

The NIALM uses pattern recognition to disaggregate the individual load from one power sensor. It offers a simple and low-cost power sensor installation; however, the accuracy is low [2]–[5]. Furthermore, since there is no additional instrument in each appliance, the NIALM system is unsuitable for individually controlling the appliance. In contrast, the IALM provides a flexible system to control the appliance individually.

Recently, the IALM system adopted IoT technology for easy and flexible accessibility. In [6], a low-cost power monitoring system based on IoT was developed to monitor the power consumption of home appliances. The ACS712 current sensor was used to measure the load's current. The ESP8266 was used to interface with the current sensor and send the data to the AWS cloud server via WiFi communication.

In [7], a multi-meter monitoring system was used to classify sixteen different loads using a machine learning technique implemented on a Raspberry Pi module. The developed Long Short-Term Memory (LSTM) network was trained using the Plug-Load Appliance Identification Database (PLAID) [8]. In the system, the smart meter sent data to the Raspberry through a Message Queuing Telemetry Transport (MQTT) protocol, which was then sent to the Google Firestore cloud server.

In [9], the IALM was used as an activity recognition system to recognize the activities of daily living, such as washing the dishes after breakfast, ironing, etc. The system consisted of the application layer, middleware layer, communication network layer, and appliances layer. The application layer defined the services for the user, i.e. activity recognition, load monitoring, and energy consumption understanding. The middle layer connects the software application and IoT devices in the form of fog computing or cloud computing. The communication network layer communicates between devices using communication networks such as 5G, WiFi, Bluetooth, Z-wave, and wired technology such as Ethernet. The perception layer consisted of sensors and actuators. The appliance layer was the physical devices such as air conditioners, refrigerators, rice cookers, etc.

The IALM was used to recognize the home appliances [10], where it introduced the training and inference frameworks in the application layer. Both frameworks were facilitated with graphical interfaces for the configuration of the system, such as setting the dataset, classifier model, and feature extraction.

Instead of the IoT technology adopted for the load monitoring system, Supervisory Control and Data Acquisition (SCADA) was also employed [11]–[13]. In [11], a microcontroller-based SCADA monitoring system was developed to monitor and control household appliances. The prototype was implemented using the Arduino microcontroller connected to the VTScada software using the Open Platform Communications (OPC) protocol.

The home appliance monitoring and control system prototype in [12] consisted of the sensor and actuator unit, the Wemos microcontroller, the Raspberry Pi single-board computer, and the IGSS SCADA software. It used the MQTT protocol, with the MQTT broker installed on the Raspberry Pi. The SCADA system for monitoring the energy consumption of household appliances was implemented in a real house [13]. The system consisted of the Programmable

Logic Controller (PLC), energy meters, current transformers, Modbus gateway, and SCADA server.

From the previous discussions, there are two approaches for load monitoring systems: IoT technology and the SCADA system. The limitations of existing systems are as follows. The IoT-based load monitoring system usually has a minimal or standard user interface dashboard. Meanwhile, the SCADA-based load monitoring system has a superior user interface; however, it still needs to be supported by IoT protocols.

In this paper, to overcome the above drawbacks, we propose the IoT-based SCADA system for monitoring the energy consumption of home appliances. The main contributions of our work are:

- It combines the most benefits of IoT and SCADA technologies, such as the powerful user interface for monitoring and supervisory systems and the scalability of IoT applications.
- It utilizes a small-size Human Machine Interface (HMI) platform provided by the Haiwell HMI, rather than using the personal computer for running the SCADA software [11]–[13]. Thus, it is suitable for installation and implementation in residential homes.
- It employs the popular Modbus protocol, measurement devices, and communication modules, which are available in the market and can be assembled and developed quickly.

The rest of the paper is organized as follows. Section 1 presents the proposed system. Section 2 describes the simulation results and discussion. Section 3 presents the conclusion.

II. SYSTEM CONFIGURATION

A. System Architecture

Fig. 1 illustrates the architecture of the load monitoring system using IoT-based SCADA. It comprises home appliances, power meters, communication modules, and the SCADA HMI. The SCADA HMI is the central unit of a SCADA system that performs remote devices' supervisory, controlling, and monitoring functions. It is noted here that we use the SCADA system for monitoring purposes only in this work. The work could be extended to cope with the control and supervisory functions accordingly.

In this work, only four appliances are considered, i.e., fan, washing machine, rice cooker, and refrigerator, even though they can be expanded to other appliances. The power meters measure appliances' power consumption and send it to the HMI through WiFi communication modules. The measured power is displayed and stored in the HMI unit that can be accessed remotely via the internet using a personal computer or smartphone.

The main unit of our proposed SCADA is the Haiwell HMI. It is an embedded, small-size SCADA server equipped with IoT technology. As the SCADA system, it provides the popular Modbus RTU and Modbus TCP communication protocols for interfacing with the remote terminal unit.

The measurement devices consist of a power meter for measuring load power consumption and a temperature (and

humidity) sensor for measuring ambient temperature and humidity. The modules are ready-used and widely available on the market, namely the SPM91 for the power meter module and SHT20 for the temperature and humidity sensor module. Thus, they provide accessible and flexible deployment. Both modules provide the Modbus protocol, thus making them suitable for SCADA communication.

WiFi communication is adopted between the measurement modules and the SCADA HMI. Since the SPM91 and SHT20 modules do not have the WiFi interface, a serial 485 Modbus to WiFi converter is required. Fortunately, the Protoss-PW11 module accomplishes this function properly.

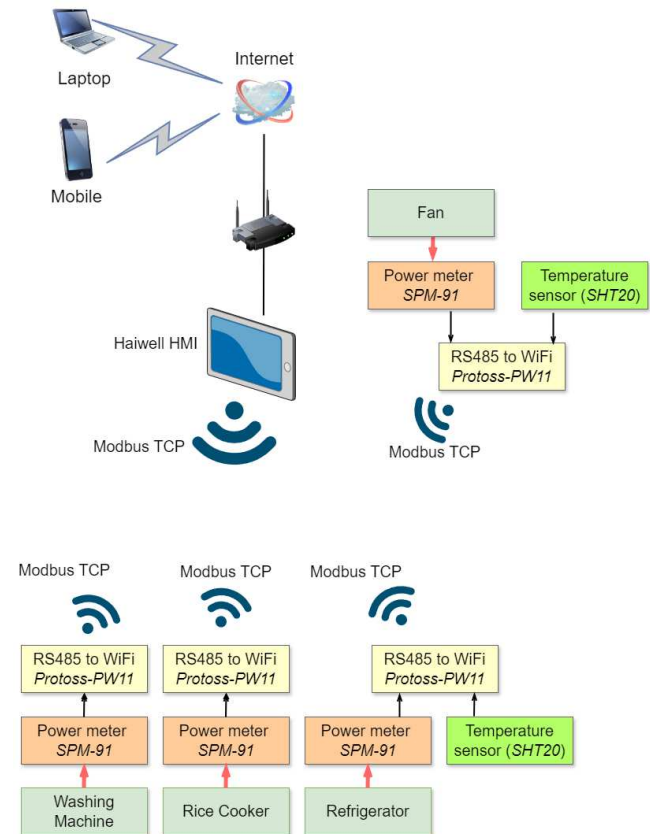


Fig. 1. Architecture of load monitoring system using IoT-based SCADA.

B. Hardware Configuration

Fig. 2 depicts the schematic diagram of the measurement device, which includes the temperature and humidity sensor (SHT20), power meter (SPM91), and RS485 Modbus to WiFi converter (Protoss-PW11). The wiring of the components consists simply and straight away of the RS485 serial communication, DC power, and AC line wiring.

Tables 1 to 3 give the specifications of SHT20, SPM91, and Protoss-PW11 measurement devices, respectively. The tables show that the devices are highly precise and accurate. Furthermore, they provide the Modbus protocol for SCADA communication and DIN rail installation for easy assembly. Fig. 3 shows a photograph of the measurement devices. The arrangement and installation of the devices are prevalent and thus can be conducted quickly.

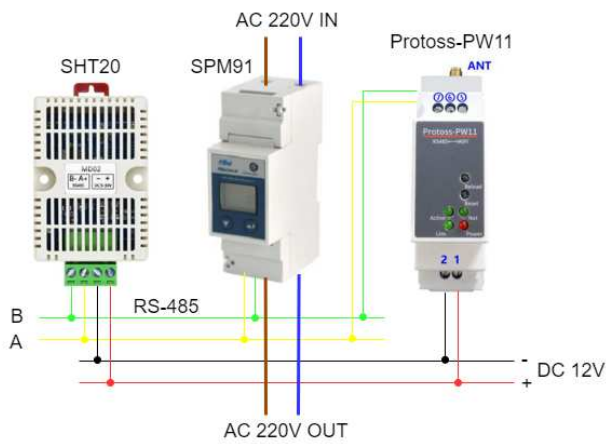


Fig. 2. Schematic diagram of measurement device.



Fig. 3. Photograph of measurement device.

TABLE I. TECHNICAL SPECIFICATION OF SHT20

<i>Parameter</i>	<i>Specification</i>
Temperature range	-40 to 60 °C
Temperature precision	+/- 0.5 °C
Temperature resolution	0.1 °C
Humidity range	0 to 80 %RH
Humidity precision	+/- 3 %RH
Humidity resolution	0.1 %RH
Power supply	5 to 30 VDC
Communication protocol	RS485 Modbus RTU
Installation	Standard 35mm DIN rail
Size (Dimension)	65 × 46 × 28.5 mm

TABLE II. TECHNICAL SPECIFICATION OF SPM91

<i>Parameter</i>	<i>Specification</i>
Rated voltage	230 VAC
Rated (Max.) current	5(63) A
Power supply	Self-supply 230V, (184V-275V)
Accuracy	Class 1 (IEC62053-21)
Communication protocol	RS485 Modbus RTU
Installation	Standard 35mm DIN rail
Size (Dimension)	36 × 100 × 65 mm

TABLE III. TECHNICAL SPECIFICATION OF PROTOSS-PW11

<i>Parameter</i>	<i>Specification</i>
Processor/Frequency	160MHz
Operating System	mbed
Network Protocol	IP, TCP, UDP, DHCP, DNS, HTTP Server/Client, ARP, BOOTP, AutoIP, ICMP, Web socket, Telnet, uPNP, NTP, Modbus TCP
WiFi interface	802.11 b/g/n, 2.412GHz-2.484GHz
Network mode	STA/AP/STA+AP
Serial port	RS485
Power supply	9 to 48 VDC@1A
Installation	C45 DIN rail
Size (Dimension)	102.03 × 64.95 × 27.50 mm

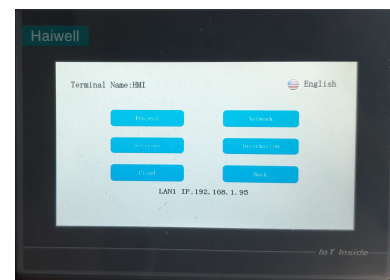


Fig. 4. Photograph of Haiwell HMI device.

TABLE IV. TECHNICAL SPECIFICATION OF HAIWELL HMI

<i>Parameter</i>	<i>Specification</i>
Programming software	Haiwell Cloud SCADA
Display screen	7 inch TFT (Touchscreen)
Display resolution	800 × 480 pixels
Storage (Flash/RAM)	2GB/512M
I/O port	Serial RS232/485, Ethernet, USB, WiFi
Power supply	24 VDC
Size (Dimension)	200 × 146 × 37 mm

The technical specification of the Haiwell HMI device is given in Table 4, and its photograph is depicted in Fig. 4. The HMI device is a 7-inch computer system with a touchscreen display. It is programmed using the Haiwell Cloud SCADA and equipped with several I/O ports: Serial RS232/485, Ethernet, USB, and WiFi. These features offer a flexible IoT-based SCADA system for monitoring home appliance power consumption.

C. SCADA Software Configuration

The Haiwell HMI is programmed with the Haiwell Cloud SCADA Software, which can be downloaded for free. Fig. 5 illustrates the software's main menu display, from which we can configure the devices and variables (tags) and build the dashboard display.

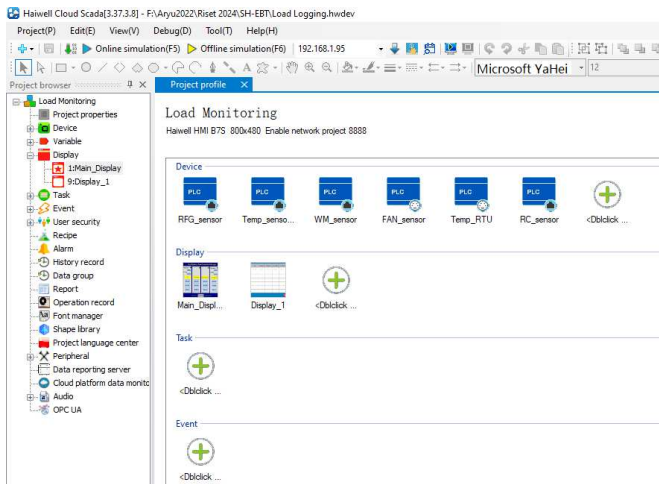


Fig. 5. Main menu display of Haiwell Cloud SCADA software.

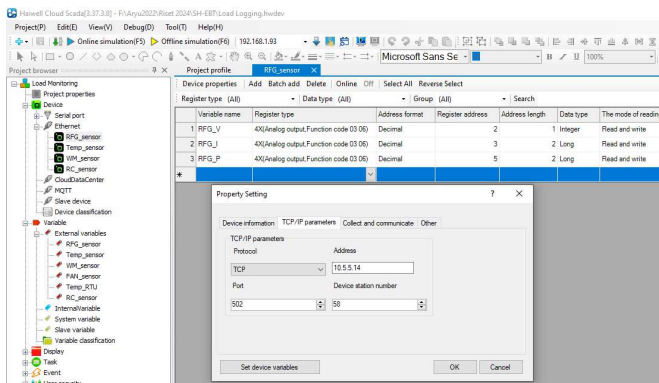


Fig. 6. Configuration of device and variable on Haiwell Cloud SCADA software.

In this work, the Modbus TCP protocol is employed to connect SCADA software with the measuring devices via WiFi communication as shown in Fig. 1. An example of the device and variable configuration in the Haiwell SCADA software is shown in Fig. 6, where we should configure the IP address of the device (in this case is the Profoss-PW11) and the function codes and the register addresses of the accessed variables.

III. EXPERIMENTAL RESULTS AND DISCUSSION

The proposed system is tested to monitor home appliances in real-time. The Haiwell HMI device is placed at home and runs continuously for a week. The measurement devices shown in Fig. 3 are placed near the appliances, which consist of the refrigerator, washing machine, rice cooker, and fan. The HMI collects data from the measurement devices every minute. The objectives of the experiments are twofold. The first is to evaluate the reliability of the SCADA system for monitoring, collecting, and storing the power consumption of home appliances. The second is to analyze the profiles of home appliances' power consumption.

Fig. 7 depicts the main display of the SCADA monitoring system, which shows the electrical parameters such as voltage, current, and power of each appliance. In addition, the temperature and humidity are also shown. The light green circle indicates that the corresponding measurement device is switched on and connected to the HMI SCADA. In contrast,

the dark green circle indicates that the measurement device is disconnected. As shown in the figure, the measurement device in the rice cooker is connected, but the current and power are zero. It denotes that the rice cooker is not switched on. The Historical Data button is used to download the recorded data in CSV file format. Based on the historical data stored in the HMI, each appliance's power consumption profile is analyzed, as discussed below.

Figs. 8 and 9 show the hourly indoor house temperature and humidity profiles, respectively. The figures show the typical profiles of indoor house temperature and humidity in a tropical area, where the higher temperature (lower humidity) occurs at noon, around 09:00 to 13:00. It is noted here that there is no air conditioner in the room under measurement. As shown in Fig. 8, the temperature fluctuates about 0.5 °C. This is caused by the specification of the temperature sensor (SHT20) as given in Table 1, where the temperature precision is ± 0.5 °C. Similarly, the humidity precision of the humidity sensor of $\pm 3\%$ causes a humidity fluctuation of about 2%, as shown in Fig. 9.

Fig. 10 shows the daily profile of indoor house temperature over a week. The daily temperature pattern is almost the same, with the highest temperature occurring at noon and then decreasing until the morning of the next day.

Figs. 11 and 12 show refrigerator power consumption's hourly and daily profiles, respectively. Fig. 11 shows that the refrigerator is switched on (power is 100 Watts) for about 45 minutes and switched off (power is 0 watts) for about 30 minutes. The switch-on period is longer at noon due to the higher ambient temperature. It is interesting to note that a high spike occurs during the starting time of the switch-on period, as shown in the figure. Fig. 12 shows that the high spike occurs almost every day. This result gives valuable information about the refrigerator's characteristics and can be used for further analysis.

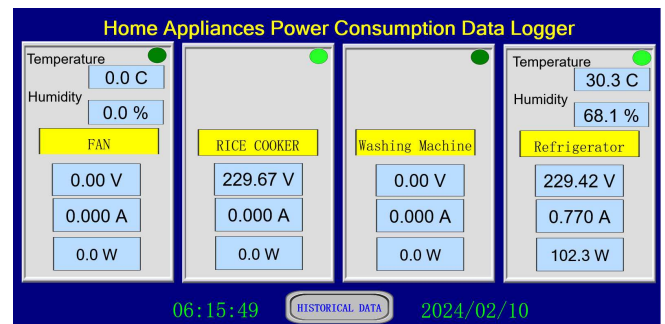


Fig. 7. Main display of the SCADA monitoring system.

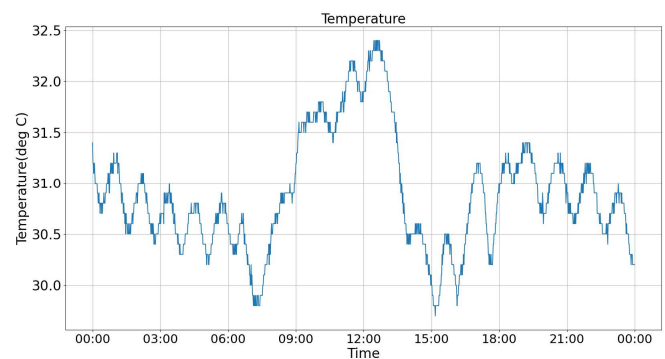


Fig. 8. Hourly profile of indoor house temperature.

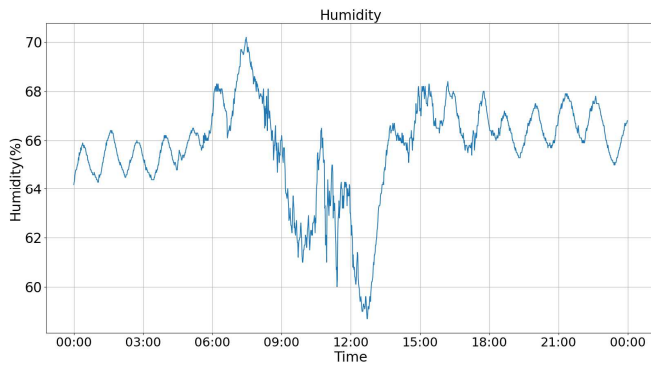


Fig. 9. Hourly profile of indoor house humidity.

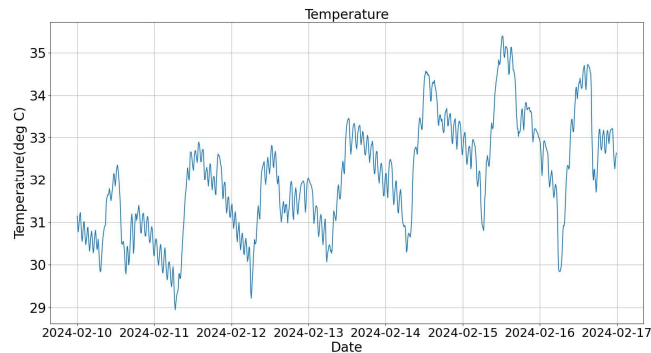


Fig. 10. Daily profile of indoor house temperature.

Fig. 13 shows the profile of washing machine power consumption. The profile shows that the washing time is about one hour, divided into a wash cycle, a rinse cycle (two times), and a spin cycle. The maximum power in each cycle is about 300 Watts. The figure suggests that the power for almost half of the washing time is minimal, thus allowing for a better understanding of its power consumption behavior.

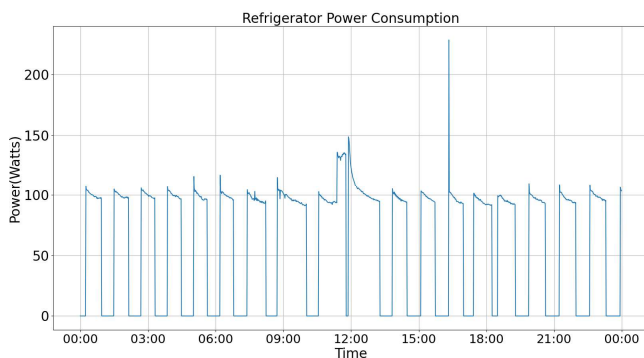


Fig. 11. Hourly profile of refrigerator power consumption.

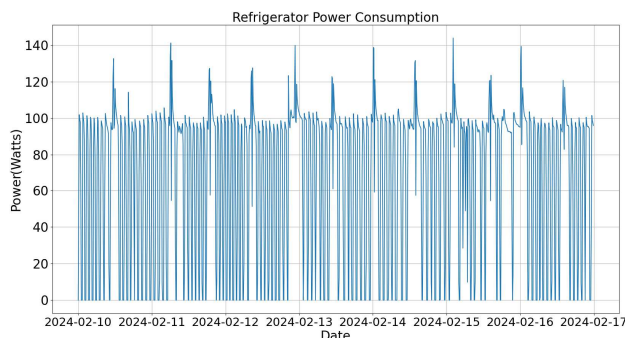


Fig. 12. Daily profile of refrigerator power consumption.

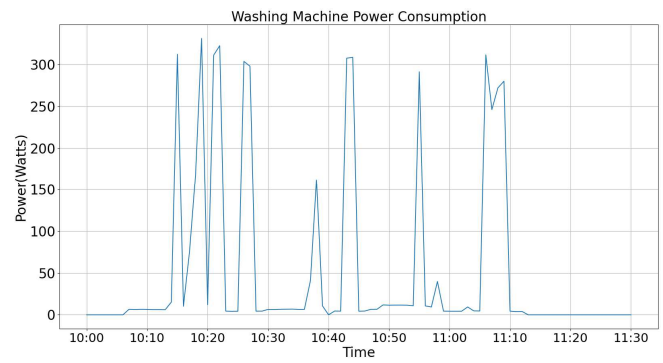


Fig. 13. Profile of washing machine power consumption.

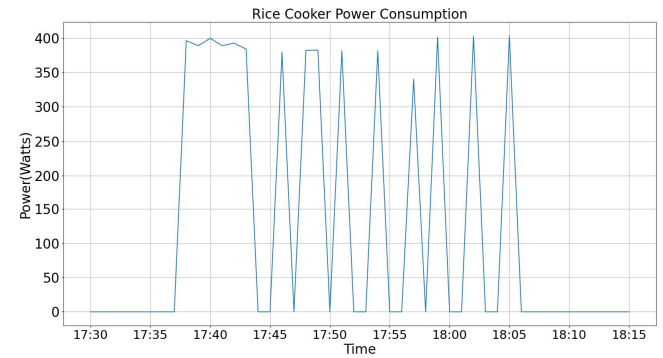


Fig. 14. Profile of rice cooker power consumption.

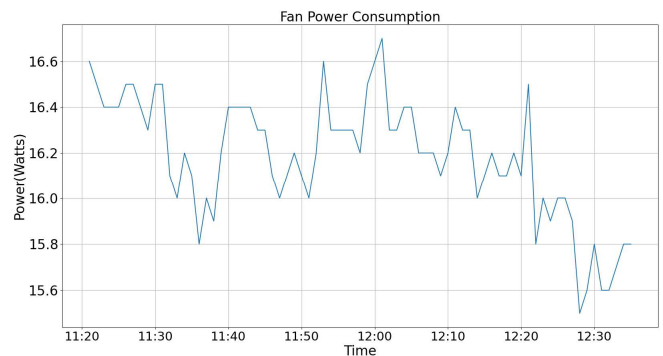


Fig. 15. Profile of fan power consumption.

Fig. 14 shows the profile of rice cooker power consumption. The figure shows that the cooking time is about 30 minutes. It consumes about 400 Watts of constant power in the first five minutes, followed by turning on and off cycles every 2 minutes. Similar to refrigerators and washing machines, examining the power consumption profile may give us more detailed information on how to design energy management systems for home appliances efficiently.

Fig. 15 shows the profile of fan power consumption. In contrast with the previous figures, the fan consumes constant power. Fig. 15 shows that due to the power measurement device and the fan's motor characteristic, the fan power consumption profile fluctuates slightly, about 0.4 Watts.

Unlike the existing system, which used specific measurement devices and a personal computer for a load monitoring data logger [8], [14], our proposed approach offers a practical solution by employing general measurement devices and a small industrial-based HMI device with IoT-enabled technology. The proposed system

could easily be upgraded for real-time energy monitoring and management.

Some limitations are found in the experiments, such as:

- The reliability of the Modbus protocol's data transfer is affected by the WiFi communication and the request-response mechanism. Sometimes, communication data transfer is lost. Fortunately, for the one-minute interval used in the experiment, there was no data transfer loss.
- The electronic module in the SPM91 power meter is energized directly by the line power; thus, the SCADA reads the voltage of the measured device as the line voltage (220V) even when the device is switched off.

IV. CONCLUSION

The load monitoring system using an IoT-based SCADA system was developed to monitor the power consumption of the home appliances. The SCADA monitoring system consisted of the SCADA master and the power measurement devices. The SCADA master application was implemented on the Haiwell HMI that can be accessed remotely via the internet. The power measurement devices that supported WiFi communication and the Modbus TCP protocol were ready-used and widely available on the market. The SCADA system was easy to assemble, develop, and implement at home for monitoring and recording the power consumption continuously with a time interval of 1 minute.

Looking ahead, our plans include the development of a larger SCADA system for load monitoring, which will be integrated with the home energy management system. This expansion will allow for more comprehensive monitoring and control. Additionally, we are preparing to conduct extensive experiments on complex home appliances, further demonstrating the versatility and adaptability of our system.

ACKNOWLEDGMENT

This work was partially supported by a Research Grant from the Ministry of Education, Culture, Research, and Technology, Republic of Indonesia, Year 2024 (No.: SP DIPA-023.17.1.690523/2024).

REFERENCES

- [1] A. Wójcik, R. Łukaszewski, R. Kowalik, and W. Winiecki, "Nonintrusive Appliance Load Monitoring: An Overview, Laboratory Test Results and Research Directions," *Sensors*, vol. 19, no. 16, pp. 1–25, 2019.

- [2] G. Bucci, F. Ciancetta, E. Fiorucci, S. Mari, and A. Fioravanti, "State of art overview of Non-Intrusive Load Monitoring applications in smart grids," *Meas. Sensors*, vol. 18, no. September, 2021, doi: 10.1016/j.measen.2021.100145.
- [3] H. Çimen, N. Bazmohammadi, A. Lashab, Y. Terriche, J. C. Vasquez, and J. M. Guerrero, "An online energy management system for AC/DC residential microgrids supported by non-intrusive load monitoring," *Appl. Energy*, vol. 307, no. July 2021, 2022, doi: 10.1016/j.apenergy.2021.118136.
- [4] C. Puente, R. Palacios, Y. González-Arechavala, and E. F. Sánchez-Úbeda, "Non-intrusive load monitoring (NILM) for energy disaggregation using soft computing techniques," *Energies*, vol. 13, no. 12, pp. 1–20, 2020, doi: 10.3390/en13123117.
- [5] S. S. Ahmad and M. H. Qutqut, "Centralized Smart Energy Monitoring System for Legacy Home Appliances Centralized Smart Energy Monitoring System for Legacy Home Appliances," 2024.
- [6] K. Luechaphonthara and A. Vijayalakshmi, "IoT based application for monitoring electricity power consumption in home appliances," *Int. J. Electr. Comput. Eng.*, vol. 9, no. 6, pp. 4988–4992, 2019, doi: 10.11591/ijece.v9i6.pp4988-4992.
- [7] Z. Solatidehkordi, J. Ramesh, A. R. Al-Ali, A. Osman, and M. Shaaban, "An IoT deep learning-based home appliances management and classification system," *Energy Reports*, vol. 9, pp. 503–509, 2023, doi: 10.1016/j.egyr.2023.01.071.
- [8] R. Medico *et al.*, "A voltage and current measurement dataset for plug load appliance identification in households," *Sci. Data*, vol. 7, no. 1, pp. 1–10, 2020, doi: 10.1038/s41597-020-0389-7.
- [9] P. Franco, J. M. Martinez, Y. C. Kim, and M. A. Ahmed, "IoT Based Approach for Load Monitoring and Activity Recognition in Smart Homes," *IEEE Access*, vol. 9, pp. 45325–45339, 2021, doi: 10.1109/ACCESS.2021.3067029.
- [10] P. Franco, J. M. Martinez, Y. C. Kim, and M. A. Ahmed, "A framework for iot based appliance recognition in smart homes," *IEEE Access*, vol. 9, pp. 133940–133960, 2021, doi: 10.1109/ACCESS.2021.3116148.
- [11] I. G. K. S. Budarsa, I. W. R. Ardana, L. G. P. Suardani, and I. P. Muliana, "Household Electricity Utilization Monitoring System Using Microcontroller Based SCADA," *J. Comput. Sci. Technol. Stud.*, vol. 4, no. 2, pp. 140–147, 2022, doi: 10.32996/jcsts.2022.4.2.16.
- [12] N. S. S. Al-Atwan and M. Nitulescu, "Monitoring and Controlling Home Appliances by Different Network Technologies," *Proc. 2020 21st Int. Carpathian Control Conf. ICC 2020*, pp. 1–6, 2020, doi: 10.1109/ICCC49264.2020.9257242.
- [13] V. Dhongade and A. D. Shaligram, "The study on consumption of household appliances with SCADA platform," *AIP Conf. Proc.*, vol. 2335, no. March, 2021, doi: 10.1063/5.0043477.
- [14] J. Kelly and W. Knottenbelt, "The UK-DALE dataset, domestic appliance-level electricity demand and whole-house demand from five UK homes," *Sci. Data*, vol. 2, pp. 1–14, 2015, doi: 10.1038/sdata.2015.7.